

EXPERIMENT 1

AIM: 1. Write a Program to implement the data link layer framing methods such as

i) Character stuffing ii) bit stuffing.

SOLUTION:

Bit stuffing is a framing technique in which each frame begins and ends with a flag (01111110). If five consecutive 1s appear in the data, a 0 is added after them so the receiver knows it is part of the data, not a flag.

Source Code:

```
#include <stdio.h>
#include <conio.h>

void main()
{
    char data[100], stuffed[200];
    int i, j = 0, count = 0;

    clrscr();

    printf("Enter bit stream (only 0s and 1s): ");
    scanf("%s", data);

    /* Check whether the input contains only 0s and 1s */
    for(i = 0; data[i] != '\0'; i++)
    {
        if(data[i] != '0' && data[i] != '1')
        {
            printf("\nInvalid input! Please enter only 0s and 1s.");
            getch();
            return;
        }
    }

    /* Perform Bit Stuffing */
    for(i = 0; data[i] != '\0'; i++)
    {
        stuffed[j++] = data[i];

        if(data[i] == '1')
            count++;
        else
            count = 0;

        /* Stuff a 0 after five consecutive 1s */
        if(count == 5)
        {
            stuffed[j++] = '0';
            count = 0;
        }
    }

    stuffed[j] = '\0';

    printf("\nOriginal Data    : %s", data);
    printf("\nBit Stuffed Data : %s", stuffed);
}
```

```
    getch();  
}
```

Output:

1.2) Character stuffing is a framing technique in which each frame begins with DLE STX and ends with DLE ETX. If DLE appears in the data, another DLE is added before it so the receiver knows it is data, not a control character.

DLE: Data link Escape

STX : Start of Text

ETX: End of Text

```
#include <stdio.h>  
#include <conio.h>  
  
#define DLE '$'  
#define STX '^'  
#define ETX '#'  
  
void main()  
{  
    char data[100], stuffed[200];  
    int i, j = 0;  
  
    clrscr();  
  
    printf("Enter Data: ");  
    scanf("%s", data);  
  
    /* Add DLE STX */  
    stuffed[j++] = DLE;  
    stuffed[j++] = STX;  
  
    /* Character Stuffing */  
    for(i = 0; data[i] != '\0'; i++)  
    {  
        if(data[i] == DLE)  
        {  
            stuffed[j++] = DLE;  
        }  
  
        stuffed[j++] = data[i];  
    }  
  
    /* Add DLE ETX */  
    stuffed[j++] = DLE;  
    stuffed[j++] = ETX;  
    stuffed[j] = '\0';  
  
    printf("\nFrame After Character Stuffing:\n\n");  
  
    for(i = 0; i < j; i++)
```

```
{
    if(stuffed[i] == DLE && stuffed[i + 1] == STX)
    {
        printf("DLE STX ");
        i++;
    }
    else if(stuffed[i] == DLE && stuffed[i + 1] == ETX)
    {
        printf("DLE ETX ");
        i++;
    }
    else if(stuffed[i] == DLE && stuffed[i + 1] == DLE)
    {
        printf("DLE DLE ");
        i++;
    }
    else
    {
        printf("%c ", stuffed[i]);
    }
}

getch();
}
```

Output: